

- 4 (a) The induced-fit hypothesis and the lock-and-key hypothesis are used to describe the mode of action of enzymes.

Explain the induced-fit hypothesis.

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- (b) Radish plants contain the enzyme peroxidase that catalyses the breakdown of hydrogen peroxide.

Students investigated the effect of increasing the concentration of hydrogen peroxide on the activity of peroxidase extracted from radish.

The results of their investigation are shown in Fig. 4.1.

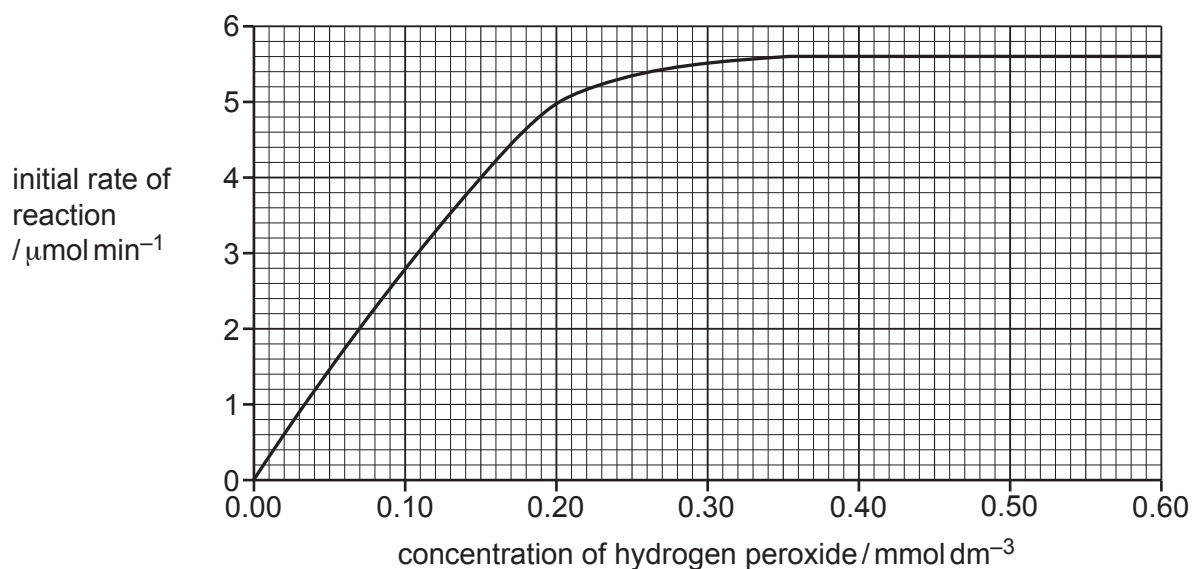


Fig. 4.1

- (i) Explain the effect of increasing the concentration of hydrogen peroxide on the initial rate of reaction as shown in Fig. 4.1.

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- (ii) The students determined the K_m for radish peroxidase as $0.10 \text{ mmol dm}^{-3}$.

With reference to Fig. 4.1, describe how they determined the K_m .

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- (c) A further investigation found that the K_m for carbonic anhydrase is 12 mmol dm^{-3} .

Describe the role of carbonic anhydrase in the transport of carbon dioxide.

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[Total: 11]